

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

SP5 - Investment and Finance

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) An asset management firm is in the process of constructing a custom equity benchmark for one of its institutional clients. The objective of the benchmark is to provide a meaningful basis against which the performance of an actively managed portfolio can be assessed. To demonstrate the impact of benchmark, CIO (Chief Investment Officer) has suggested to have a detailed calculation example for clients.

Stock	Price (In ₹)	Shares (millions)	Free float factor	Corporate Action During the Quarter
A	50	100	.80	1 for 5 rights issue at INR 40
B	80	50	.60	Paid cash dividend of INR 4 per share
C	40	150	.90	No Corporate Action

At the end of the period, the prices change to ₹55, ₹72 and ₹44 respectively.

i) Marketing team has asked for details of the calculation (using market capitalisation approach) to create marketing material for clients. You have been asked to provide the required details. [3]

ii) Following Corporate actions were taken during the quarter.

Stock	Corporate Action During the Quarter
A	1 for 5 rights issue at INR 40
B	Paid cash dividend of INR 4 per share
C	No Corporate Action

How would Corporate actions as taken, will impact the overall returns? [4]

iii) List the reason why the adjustment for free float and corporate actions are needed: [3]

iv) Equity fund manager have called out the limitations of using market capitalisation index as a benchmark for active portfolio management. What would be the key reasons for their objections? [2]

[12 Marks]

Q.2) You are an Investment Analyst who is evaluating a listed company for possible inclusion in an actively managed equity portfolio. The company operates in a cyclical sector, and has reported strong earnings growth in recent years.

The following information is available:

- Earnings per share (EPS): ₹25
- Dividend payout ratio: 40% (*assume dividends are based on this payout ratio*)
- Expected growth rate for the next 3 years: 8% per annum
- Long-term growth rate thereafter: 4% per annum
- Required rate of return: 11%

The analyst is concerned that recent performance may not be fully sustainable.

- i) Estimate the value of the share using a two-stage dividend discount model, based on the information provided. Clearly state any assumptions made. [7]
 - ii) Discuss the key assumptions underlying your valuation and comment on their reasonableness given the cyclical nature of the business. [4]
 - iii) Briefly comment on how sensitive your valuation is to changes in the assumed growth rates. [2]
- [13 Marks]**

Q. 3) DJT Ltd is a large insurance company with operations across life and general insurance business lines. The company has experienced stable premium growth over recent years and is now considering a strategic expansion involving investment in digital distribution capabilities and acquisition of an asset management platform.

The Board is reviewing the company's capital management strategy and is considering financing the expansion through either additional debt issuance or a new equity issue. Senior management believes that increasing leverage may improve shareholder returns. However, concerns have been raised regarding the impact of rising interest rates, solvency pressures and deterioration in credit quality.

The following information relates to DJT Ltd for the latest financial year.

Income Statement (INR crore)

Item	Amount
Revenue	500
EBIT	60
Interest Expense	10
Profit Before Tax	50
Tax Expense	15
Net Profit	35

Balance Sheet (INR crore)

Item	Amount
Total Assets	400
Debt	150
Equity	250

Additional information:

- Existing debt is predominantly floating-rate debt.
- Market interest rates are expected to rise by 2% over the next year.
- The company is considering raising an additional ₹100 crore to fund expansion.
- Two financing alternatives are under consideration:

Alternative 1 — Debt Financing

- Entire ₹100 crore to be raised through long-term debt.
- Interest rate on new borrowing: 11% per annum.
- Expansion expected to increase EBIT by 20%.

Alternative 2 — Equity Financing

- Entire ₹100 crore to be raised through equity issuance.
- Expansion expected to increase EBIT by 15%.
- Corporate tax rate is 30%.
- Credit rating agencies have indicated that:
 - interest coverage below 3.5 times may lead to a negative outlook;
 - debt-to-equity ratio above 1.0 may increase refinancing concerns.

i) Calculate the following ratios based on the company's current financial statements:

- Debt-to-equity ratio;
- Debt-to-assets ratio;
- Interest coverage ratio;
- Return on equity.

Comment briefly on the company's current financial position.

[5]

ii) It has been decided to go ahead with Equity financing. Assess the impact on Income statement and above ratios. [4]

iii) Chief Risk Officer wants a detailed discussion on risks associated with increasing leverage in a rising interest rate environment. You have been tasked to create a note for circulation. What all risks would you cover in the note along with key reasons for those risks. [4]

iv) If Debt financing is used, can interest rate swap could be used to manage the risk of floating rate interest? [1]

[14 Marks]

Q.4) A bank is considering securitising a pool of residential mortgages through the creation of a Special Purpose Vehicle (SPV).

The following information is available:

Item	Amount
Total value of mortgage pool	₹500 cr.
Expected annual default rate	3%
Expected annual prepayment rate	5%
Recovery rate on defaulted mortgages	60%
Average mortgage coupon	9% p.a.
Weighted average maturity	12 years

The SPV will issue the following securities:

Tranche	Size (₹ crore)	Coupon
Senior Notes	350	7%
Mezzanine Notes	100	10%
Equity Tranche	50	Residual cash flows

Additional information:

- Losses are allocated sequentially:
 - equity tranche absorbs losses first;
 - followed by mezzanine;
 - senior tranche last.
 - The originating bank intends to retain the equity tranche.
 - Market interest rates are expected to rise over the next year.
 - Regulators have recently increased scrutiny of securitisation structures and liquidity risk management practices.
- i) Head of marketing wanted to understand the above proposal better and has asked for your help in understanding
- the process of securitisation.
 - the role of the SPV in transferring risk.

Discuss the points that will be covered in your discussion. [3]

ii) Estimate the expected loss arising from defaults over next 1 year. [2]

iii) Explain how prepayment risk can affect investor returns. [3]

iv) Discuss the suitability of such securities for institutional investors, including the associated risk-return trade-offs relative to traditional asset classes and any regulatory or investment mandate constraints that may affect their use. [4]

[12 Marks]

Q.5) An investment committee of a large institutional fund is carrying out its annual review of the performance of an externally appointed active equity manager.

The manager is benchmarked against a broad market index. Over the past year, the portfolio has outperformed the benchmark. However, the committee wishes to understand the sources of this outperformance and assess whether it reflects genuine and repeatable investment skill.

The committee is particularly interested in distinguishing between the impact of asset allocation decisions and stock selection decisions, and in understanding whether the results could have been influenced by prevailing market conditions.

- i) Discuss why is attribution analysis and its components a preferred approach for performance assessment. [4]
- ii) The following information has been provided:

Sector	Portfolio Weight (%)	Benchmark Weight (%)	Portfolio Return(%)	Benchmark Return (%)
Technology	35	30	14	11
Healthcare	20	20	9	10
Financials	15	25	6	8
Energy	10	10	5	4
Consumer	20	15	11	9

Calculate the allocation and selection effects for each sector. In addition, comment briefly on whether interaction effects are likely to be significant in this case. [5]

iii) List limitations of attribution analysis when assessing the performance of an active equity manager. [3]

iv) Using your results, explain the main drivers of the portfolio's outperformance relative to the benchmark, and assess whether these drivers are consistent with the stated investment strategy of the manager and the broader portfolio objectives. [4]

v) The committee notes that the manager was overweight in Technology and underweight in Financials. Assess whether these decisions appear to have added value, and evaluate whether they are consistent with the fund's stated investment strategy and any relevant regulatory or governance constraints. [3]

vi) Evaluate the extent to which attribution analysis alone provides sufficient evidence for the committee to decide whether to retain the manager basis the risk taken, consistency of performance, suitability of the benchmark used and any governance, regulatory or reporting requirements relevant to the review process. [5]

vii) Assess whether the observed outperformance could reflect exposure to rewarded systematic risk factors, such as growth, momentum, quality or sector biases, rather than genuine manager skill. [3]

[27 Marks]

Q.6) An insurance company is in the process of transitioning ₹200 crore of its domestic equity portfolio from an actively managed strategy to a passive index-tracking strategy. The current portfolio differs materially from the target benchmark portfolio and therefore significant rebalancing trades will be required.

The following transition cost estimates have been provided:

Cost Component	Estimate
Explicit transaction costs	1.5% of portfolio value
Market impact costs	2.0% of portfolio value

Expected temporary cash holding during transition	10% of portfolio
Expected transition period	20 trading days
Expected annual equity market return	12%
Annualised equity market volatility	20%

Additional information:

- Market conditions are currently volatile.
- The company is concerned about tracking error during the transition period.
- The target passive strategy is intended to replicate a broad market-cap-weighted benchmark.
- The insurer has significant solvency capital sensitivity to equity market movements.
- The transition may either be implemented internally or outsourced to a specialist transition manager.

i) Calculate the total expected cost of the transition. [4]

ii) Explain the main risks associated with carrying out such a transition. [6]

iii) Explain how derivatives could be used to reduce these risks, and discuss any regulatory constraints and behavioural factors that may affect the effectiveness of such strategies. [2]

[12 Marks]

Q.7) An insurance company has the following balance sheet profile:

Item	Value
Market value of assets	₹1,200 crore
Market value of liabilities	₹1,050 crore
Asset duration	7 years
Liability duration	10 years

The investment team is considering changes to the asset allocation in response to recent movements in interest rates.

Additional information:

- Interest rates are expected to decline over the next 12 months.
- The company currently invests primarily in fixed income securities.
- Management is concerned about the impact of interest rate movements on the company's solvency position.
- The insurer is subject to regulatory solvency capital requirements based partly on interest rate stresses.
- The investment team is considering:
 - increasing allocation to long-duration bonds;
 - using interest rate swaps;

- and increasing exposure to equities.
- i) Calculate the duration gap for the insurer and its implication on the insurer. [4]
- ii) Estimate the relative change in the surplus if interest rates fall by 1%. Comment on your answer. [3]
- iii) Discuss how behavioural biases may influence investment decisions in this context, particularly in response to recent market movements. [3]

[10 Marks]
